Difference decomposition and the additive-quadruple estimate

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1 Inputs and conventions

We use the natural-ultrapower convention of GTZ Appendix A and the project shift theorem in tautification-003. Its acceptance remains separate. A bound $\gg 1$ is positive and standard. Write N for an unbounded limit integer, $J = [N]$ and $I = [-N, N] \cap {}^*\mathbb{Z}$. All changes of variables are integer identities at the representative indices. The author erratum guides the proof. Its Proposition 3.1 and modified Proposition 7.3 are the conclusions here, rather than assumed premises.

The shift theorem gives a lower-support sequence with fixed geometry and uniform Lipschitz bound. It gives neither a separated finite decomposition nor a correlation conclusion. Products, conjugates, and translates use product manifolds, conjugated tests, and translated polynomial orbits, respectively. A bounded internal scalar depending only on parameters is degree zero in the averaging variable.

2 Difference decomposition

Proposition 1 (LP02)**.** *For standard $s \geq 3$, let χ be a multidegree $(1, s-1)$ nilcharacter. For every standard $\epsilon > 0$ each component of $D_k(h, n) = \chi(h+k, n) \otimes \overline{\chi(h, n)}$ is approximated uniformly in k, h, n to error ϵ by*

$$\sum_{j=1}^{L} \Psi_{j,k,a_j}(n) B_{j,k}(h, n).$$

Here L is standard, Ψ_{j,k,a_j} is a component of a unit-vector degree-$(s-1)$ nilcharacter, and $B_{j,k}$ is supported on the downset below $(1, s-2)$. The families are internal with standard finite geometry lists and uniform bounds.

Proof. Apply the project shift theorem, Theorem 3, with positive index $(1, 0)$ and shift $(k, 0)$. The support is in the union of the downsets $J_0 = \downarrow (0, s-1)$ and $J_1 = \downarrow (1, s-2)$. We now give the fixed geometric split explicitly. Write G/Γ for the taut lower-support representation from the shift theorem and $J = J_0 \cup J_1$. For each positive $a \in J$ take a rational basis of $\mathfrak{g}_a$ for the rational structure of Γ. This exists because G_a is rational. Take a free rational Lie algebra with one generator of weight a for each basis vector. Kill all homogeneous words whose total weight is outside J. The excluded weights form an upward-closed set, so these words generate a homogeneous ideal. All generator weights are positive. Thus word length is bounded by $\max_{a \in J} |a| = s - 1$ in the quotient, which is finite-dimensional and nilpotent. Let $\mathfrak{u} = \bigoplus_{a \in J \setminus \{0\}} \mathfrak{u}[a]$ denote it and let U be its simply connected real group. Set $\mathfrak{u}_b = \bigoplus_{a \geq b} \mathfrak{u}[a]$. This is a descending rational multidegree filtration with the required bracket rule.

Sending generators to their basis vectors defines a rational homomorphism $\rho : U \to G$. The killed words map to zero by the filtration rule in G. It maps each positive layer onto G_b: the

generators tagged b span $\mathfrak{g}_b$, and all weights at least b map into that layer. At zero it is onto because G is taut and generated by its coordinate groups $G_{(1,0)}$ and $G_{(0,1)}$. Thus the polynomial lifts exist by published GTZ Corollary B.10, applied to this layerwise-surjective filtered map, and can be selected internally at the representative indices.

Choose a homogeneous rational basis of $\mathfrak{u}$ and let L_0 be its integer span. There is a positive integer D such that $\Lambda = \exp(DL_0)$ is a lattice and $\rho(\Lambda) \subset \Gamma$. Here are the scaling details. In this basis BCH is a finite rational polynomial with linear part addition. For two inputs in DL_0, terms of degree at least two lie in DL_0 once D clears all the coefficient denominators. Inversion negates the logarithm. Thus this set is a subgroup, discrete in exponential coordinates. It is cocompact: order the basis by increasing total weight and successively reduce each weight's logarithm coordinates modulo D. Right multiplication by an element of the corresponding homogeneous lattice leaves lower weights unchanged, and at the current weight is ordinary addition. The resulting bounded box of logarithms is a compact set of representatives. The same argument applies to every filtration layer and every homogeneous quotient. To impose $\rho(\Lambda) \subset \Gamma$, use rational Mal'cev coordinates of the second kind adapted to Γ, as in GTZ Section 3. The coordinates of $\rho(\exp x)$ are finite rational polynomials in the chosen coordinates of x, with zero constant term. Taking D divisible by their denominators makes the coordinates of $\rho(\exp(Dx))$ integral for $x \in L_0$. This gives the required lattice inclusion. Increasing D preserves the BCH and compactness arguments.

For $i = 0, 1$, quotient U by the connected subgroup with Lie algebra the sum of homogeneous weights outside J_i. These sums are ideals by the downset property. The image lattices are the exponential lattices in the surviving homogeneous bases. The pair of quotient maps separates points of U/Λ: if two points have the same image in both, the logarithm of their group difference lies in the projected lattice in each quotient. Each homogeneous weight survives in at least one quotient. Hence every coordinate of that logarithm belongs to $D\mathbb{Z}$, so the difference lies in Λ. This proves separation on the quotient spaces, rather than just intersection-triviality of the two ideals.

The pullbacks of Lipschitz tests from the two compact quotients generate a conjugation-closed algebra containing constants and separating points. Stone–Weierstrass gives a uniform approximation by finite sums of products of these tests. The lifted tests of the shift family have a common Lipschitz and sup bound on this fixed compact manifold, since the map to G/Γ is fixed and smooth. A standard finite $\epsilon/4$ net for this bounded equicontinuous class gives finitely many standard product lists, each chosen to tolerance $\epsilon/4$. Select the first net entry within $\epsilon/4$ using an internal sup-norm comparison at representative indices. Use a smaller net radius if needed. Thus the splitting error is at most $\epsilon/2$, uniformly in k, h, n. Pad the lists with zero factors to a standard common length.

In the J_0 quotient all surviving generators have zero first weight and strictly positive second weight. The restricted one-variable degree filtration is therefore taut: its degree-zero and degree-one groups coincide. The highest layer is central by the bracket law and the degree bound. The orbit is independent of h, since the first-variable derivative lies in the trivial first-coordinate group. Restrict to $h = 0$ and apply published Lemma E.5 with this taut one-variable geometry of degree $s - 1$. Its test decomposition gives linear functionals of unit-vector nilcharacters times degree-at-most-$(s-2)$ tests. Expand the functionals in coordinates and absorb their coefficients and lower-degree factors into the J_1 factors. The product still has support J_1. Choose individual Fourier tolerances after the finite lists and their sup bounds so that the added error is at most $\epsilon/2$. On fixing h, k, the lower factor has degree at most $s - 2$ in n by the $(0, 1)$ derivative index. All geometry lists are fixed and finite, and all orbit data and choices are internal. □

3 The two averages

Proposition 2 (LP03)**.** *Under the hypotheses of GTZ Theorem 7.2, for standard $s \geq 3$, there are $\gg N^3$ additive quadruples in H for which*

$$\chi_{h_1}(n) \otimes \chi_{h_2}(n + h_1 - h_4) \otimes \overline{\chi_{h_3}(n)} \otimes \overline{\chi_{h_4}(n + h_1 - h_4)}$$

is $(s-2)$-biased on J. The witnessing tests have standard bounded complexity and positive standard correlation.

Proof. Extend f by zero outside J. GTZ A.11 and A.12 select the lower-degree correlation tests internally on the dense internal set $H \subset I$. The relation includes a selected positive lower bound $1/j$ with standard integer j. Select global representatives of the lower-degree tests and of the restricted nilcharacters by the same internal selection, storing the representations as part of the witness. Only their original values on J enter the unsmoothed identities. The bounded-complexity descriptions are a countable union of internal sets. The boundedness principle A.5 gives uniform standard complexity and a common positive lower bound. Expand the bounded-dimensional tensor contraction in coordinates. Retain a positive-density subset of H with a fixed witnessing coordinate pair by the standard finite pigeonhole. Write U_h, V_h for the conjugates of those coordinates of $\chi(h, \cdot)$ and χ_h. Then there are scalar lower-degree tests c_h such that

$$\left|\mathbb{E}_{n \in J} f(n + h)\overline{f(n)}U_h(n)V_h(n)c_h(n)\right| \geq \eta > 0$$

on the retained set, with standard uniform bounds. Align the phase by multiplying c_h by a unit scalar. Set $c_h = 0$ outside that set and extend V_h boundedly there. This is an internal choice. The real average over $h \in I$ is $\gg 1$.

Apply Cauchy–Schwarz in n to eliminate $\overline{f(n)}$. It gives

$$\mathbb{E}_{n \in J} \left|\mathbb{E}_{h \in I} f(n + h)U_h(n)V_h(n)c_h(n)\right|^2 \gg 1.$$

Expand the square and put $h' = h + k$, with $K = [-2N, 2N] \cap {}^*\mathbb{Z}$. The exact new parameter cutoff is $1_{h+k \in I}$, degree zero in n. On $I \times K$ the new normalized average is $|I|/|K|$ times the old one. With $c_{h,k} = 1_{h+k \in I} c_h \overline{c_{h+k}}$ this gives

$$\left|\mathbb{E}_{h \in I, k \in K, n \in J} f(n + h)\overline{f(n + h + k)}U_h(n)\overline{U_{h+k}(n)}V_h(n)\overline{V_{h+k}(n)}c_{h,k}(n)\right| \gg 1.$$

The U product is a component of D_k. Take LP02 with error less than half the correlation divided by a common bound for the other factors. A term of the standard finite sum retains a positive standard correlation. Absorb its lower factor into $c_{h,k}$.

For its surviving coordinate $\Psi_{k,a}(n)$ use the identity

$$\Psi_{k,a}(n) = \sum_b [\Psi_{k,a}(n)\overline{\Psi_{k,b}(n + h)}]\Psi_{k,b}(n + h).$$

The identity uses the unit vector norm of $\Psi_k(n+h)$. Individual coordinates need not have modulus one. The bracket is a component of the conjugate transpose of the shift tensor. The one-variable degree case of the project shift theorem gives degree at most $s-2$, uniformly and internally in h, k, using the finite geometries of LP02. Select a surviving term by the same finite averaging argument. Absorb the bracket into $d_{h,k}$ and write the remaining coordinate as $P_k(n + h)$. The lower test is still exactly zero unless both $h, h + k$ are retained shifts.

Set $m = n + h$ and $M = [-2N, 2N] \cap {}^*\mathbb{Z}$. The change to the full M average, with the exact cutoff, multiplies the normalized correlation by $N/|M|$. We have

$$\left|\mathbb{E}_{m\in M,k\in K} f(m)\overline{f(m+k)}P_k(m)\mathbb{E}_{h\in I}1_J(m-h)V_h(m-h)\overline{V_{h+k}(m-h)}d_{h,k}(m-h)\right| \gg 1.$$

The outer factor depends only on (m, k) and is bounded. The second Cauchy–Schwarz in (m, k) gives $\gg 1$ for the mean squared modulus of the inner average. Expand with independent $h, h' \in I$. Put

$$(h_1, h_2, h_3, h_4) = (h, h' + k, h + k, h'), \qquad n = m - h.$$

The shift substitution is injective, with inverse $(h, h', k) = (h_1, h_4, h_3 - h_1)$. It has $h_1 + h_2 = h_3 + h_4$. The first cutoff restricts n to J, where $m = n + h \in M$ automatically. The second is $1_J(n + h_1 - h_4)$. The four factors in the new average are

$$V_{h_1}(n)V_{h_2}(n + h_1 - h_4)\overline{V_{h_3}(n)V_{h_4}(n + h_1 - h_4)}.$$

The lower factor is $d_{h,k}(n)\overline{d_{h',k}(n + h_1 - h_4)}$ times the remaining cutoff. The ratio $N/|M|$ is comparable to one. The parameter vanishing is exact unless all four shifts are retained. This proof keeps zero extension throughout.

To absorb the cutoff, choose fixed $0 < \lambda < 1/4$. On the circle $\mathbb{R}/16\mathbb{Z}$ take a piecewise linear test equal to one on $[\lambda, 1 - \lambda]$, zero off $[-\lambda, 1 + \lambda]$ in $[-8, 8)$, between zero and one, with Lipschitz bound $O(1/\lambda)$. Evaluate it at $(n + h_1 - h_4)/N$. This is a one-step nilsequence in n, with fixed geometry and internal orbit. Its argument lies in $[-2, 3]$ and does not wrap. The normalized mean absolute error against the cutoff is $O(\lambda + 1/N)$ uniformly in all shifts: only two endpoint bands of total $O(\lambda N + 1)$ integers differ. Multiplying by bounded factors changes only the constant. Choose λ after the positive correlation bound so the error loses at most half. Since $s \geq 3$, the one-step test is absorbed into degree at most $s - 2$. Parameter vanishing persists.

If the outer correlation has modulus at least $\beta > 0$ and each inner average has modulus at most B, at least a fraction $\beta/(2B)$ of the inner averages have modulus at least $\beta/2$. The shift triple box has size comparable to N^3. By injectivity, these yield $\gg N^3$ additive quadruples in H. Conjugating the scalar correlations undoes the initial choice of V_h and preserves lower degree. The resulting scalar correlations are coordinates of the claimed tensor, giving bias on the original interval J. □

4 Scope

The published proof inputs used here are GTZ Appendix A, Corollary B.10, E.5, the correlation conventions, and the stated hypotheses of Theorem 7.2. The project shift input and these exact applications await separate nonauthor review. This unit supplies the additive-quadruple result, not the subsequent frequency analysis, the other affected symbol rules, or the final GI theorem.